

Integrated Pick Scheduling and Conveyor Speed Control for a Delta-Robot Sorting Cell

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Receive:	This study develops an online algorithm that jointly schedules picks and controls conveyor speed for a delta-robot sorting system. Each part on the conveyor stays within the robot's reachable area only for a limited time; if it is not picked before leaving, it is lost. The goal is to pick as many parts as possible before time runs out – an online scheduling problem with time-varying release times and deadlines, $1 r_j(t), d_j(t), \text{online} \sum U_j$. These deadlines are not fixed: they depend on the conveyor speed, which the controller can adjust. Slowing the belt gives more time for the part being chased, but delays every part behind it. The algorithm treats scheduling and conveyor speed as one problem, using a two-loop structure. Because the camera sits upstream, the inner loop sees each part before it arrives, decides which part to pick next, and predicts how many parts on the belt will likely be missed. The outer loop uses that prediction to adjust the belt speed. The time to switch between two picks depends only on the type of the part just picked, so it is folded into the pick time. The algorithm is validated on a self-built delta-robot cell – vacuum gripper, rotation axis, encoder-equipped conveyor, a YOLO-OBB vision model, two PLCs, and a control computer – running at 30-60 picks per minute.
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